Supplementary Notes for Security Self-Ablation

1 Experimental Setup

The security self-ablations isolate two mechanisms, AMM pricing and on-chain payment enforcement, under adversarial participation. Each run uses 200 buyers, 100 liquidity providers, and 30 tokens per resource, and varies the malicious-participant ratio from 0% to 30%. Three scenarios are considered. In seller collusion, malicious sellers raise asks by 1.2–1.5× and reduce their listing rate to 30%, and the full RWA is compared with RWA without AMM. In buyer collusion, malicious buyers lower bids to 60–80% and reduce spending per step to 30%, under the same AMM switch. In payment default, malicious buyers attempt nonpayment with probability 0.5, and on-chain enforcement is switched on or off while AMM is kept on. The common metric is paid resource allocation divided by initial physical capacity.

2 Results and Analysis

As shown in `ablation_robustness.pdf`, three findings emerge. First, removing AMM under seller collusion lowers paid utilization, but the degradation is a slowdown rather than a permanent deletion of capacity. Colluding sellers can raise prices and withhold listings, yet the AMM pool continues to quote from reserves, so capacity is eventually released at pool-determined terms rather than at the colluders' asks. The AMM therefore acts as a backstop against strategic withdrawal from the order book.

Second, removing AMM under buyer collusion produces a comparable pattern. Without AMM, lowball bids succeed only when they cover asks, and trade occurs at the midpoint; with AMM, quotes follow reserves, so submitted bid shading cannot directly set the execution price. Both variants share the volume reduction induced by lower spending per step, but the AMM variant prevents colluding buyers from translating bid shading into a disproportionate share of paid allocation.

Third, the payment-default panel is the sharpest ablation. Without on-chain enforcement, unpaid delivery can occur and is excluded from successful allocation, so paid utilization falls as the malicious ratio rises. Enforcement reverses this by prelocking 10% of initial buyer cash: a defaulter's payment is recovered from collateral, a 10% trade-value penalty is applied, and the defaulter is excluded, while insufficient collateral rejects delivery outright. Default therefore does not convert into unpaid consumption of capacity. Taken together, AMM pricing supplies pricing resilience while on-chain enforcement supplies settlement finality, and neither substitutes for the other; the security argument for the full RWA design is compositional rather than resting on a single component.